openMPPT Hardware Design Calculations

Theoretical basis and calculations for component selection

1. Power Stage Parameters

Parameter	Symbol	Value	Notes
Switching Frequency	f_{sw}	100 kHz	Defined in `mppt.h` (`TIMER_PERIOD = 240`)
Maximum Input Voltage	$V_{in,max}$	80 V	Hardware limit
Maximum Output Current	$I_{out,max}$	20 A	Safety limit
Nominal System Voltage	V_{sys}	12V / 24V / 48V	Target batteries
Gate Drive Voltage	V_{drive}	10 V	Optimized for IRS21867 thermals (v1.3)

Planned change: 100kHz → 200kHz main switching frequency. Not yet implemented — `TIMER_PERIOD` in `mppt.h` (currently 240) and the table above still reflect 100kHz. `setup_cubemx_env_auto.py` already derives `TIMER_PERIOD` from `MPPT.ioc` and `main.c`'s PWM limits/dead-bands are dynamic, so the firmware side should follow from changing the timer config. See Section 8 for the full component re-check.

2. Gate Drive & Auxiliary Power Optimization (v1.3)

Primary Aux Supply (10V) - SCT2A25: Replaces the unstable 12V XL7005A with a robust 100V-rated step-down (SCT2A25).

- Output is tuned to **10V** instead of 12V.
- **Feedback Divider**: $V_{FB} = 1.2\text{V}$
  - $V_{OUT} = 1.2 \times \left(1 + \frac{R_{up}}{R_{down}}\right)$
  - Using E96 series: $R_{up} = 110\ \text{k}\Omega$, $R_{down} = 15\ \text{k}\Omega \rightarrow V_{OUT} = 1.2 \times \left(1 + \frac{110}{15}\right) = 10.0\text{V}$
- **Switching Frequency**: Fixed at 300kHz.
- **Inductor** (L_1): For 10V out, a standard **33µH or 47µH** inductor (e.g., 6x6mm shielded) is suitable.
- **UVLO Divider**: To set start at 12.5V: $R_{UVLO\_TOP} = 430\ \text{k}\Omega$, $R_{UVLO\_BOT} = 47\ \text{k}\Omega$ (Both are E24).
- **Bootstrap Diode**: **US1M** (1000V, 1A Ultrafast, $t_{rr} \leq 75\text{ns}$).

Secondary Aux Supply (3.3V Logic) - SY8120:

- Uses a "Cascaded Buck" architecture: 10V → Tiny Sync Buck (SY8120) → 3.3V.
- **Feedback Divider**: $V_{FB} = 0.6\text{V}$
  - $V_{OUT} = 0.6 \times \left(1 + \frac{R_{up}}{R_{down}}\right)$
  - Using E24 series: $R_{up} = 68\ \text{k}\Omega$, $R_{down} = 15\ \text{k}\Omega \rightarrow V_{OUT} = 0.6 \times \left(1 + \frac{68}{15}\right) = 3.32\text{V} \approx 3.3\text{V}$
- **Switching Frequency**: Fixed at 500kHz.
- **Inductor** (L): **4.7µH** (recommended for 3.3V out).

3. Transient Voltage Suppression (TVS) Strategy

Primary Bus Protection (80V):

- **Component**: **5.0SMDJ85CA** (Bidirectional, 5kW).
- **Rationale**: Standoff (V_{RWM}) of 85V ensures no leakage at 80V. 5kW rating handles massive inductive kickback and solar surges.
- **Placement**: Immediately adjacent to VIN/VOUT XT60 connectors.

Gate Drive Protection (10V):

- **Component**: **H12VH22U** (6kW Surge).
- **Rationale**: 12V standoff ensures invisibility at 10V rail. Protects sensitive IRS21867 drivers (25V max) from regulator failure.

Logic Protection (3.3V):
- **Component**: **H3V3L06B** (ESD).
- **Rationale**: 3.3V standoff is required because STM32 Absolute Max is only 4.0V. Provides last line of defense against ESD and noise.

4. Inductor Sizing (Main Power Stage)

Targeting a peak-to-peak ripple current (ΔI_L) of 20% of $I_{\text{out,max}}$ (**4.0 A**).

Buck Mode Worst-Case: (50% Duty Cycle, e.g., 80V in, 40V out)

$$L_{\text{buck}} = \frac{V_{\text{out}} \times (V_{\text{in}} - V_{\text{out}})}{\Delta I_L \times f_{\text{sw}} \times V_{\text{in}}}$$

At $f_{\text{sw}} = 100\text{kHz}$: $L_{\text{buck}} = \frac{40\times40}{4.0\times100000\times80} = *50.0\mu\text{H}*$

v1.3 Component Selection: Target inductor size: **33 μH to 47 μH**.
- **CRITICAL**: Saturation Current (I_{sat}) **MUST be > 25A**. Using a lower I_{sat} (like 11A) will cause magnetic collapse and MOSFET destruction.

5. Bulk Capacitor Selection

Targeting an output voltage ripple (ΔV_{out}) of **< 100mV**.

$$\text{ESR}_{\text{max}} = \frac{\Delta V_{\text{out}}}{\Delta I_L} = \frac{0.1\text{V}}{4.0\text{A}} = *25m\Omega*$$

v1.3 Capacitor Strategy:
- Use **4x 330μF 100V Low-ESR Electrolytic** (e.g., Ymin LKML series) in parallel.
- **Total Bank ESR**: $\approx 11.75m\Omega$ (Passes 25mΩ limit).
- **Total Ripple Capacity**: ≈ 8.5 A (Safe for 20A operation).

6. Voltage Divider & ADC Scaling

Voltage Sensing LPF (v1.3 Optimized):
- $R_{\text{top}} = 200\ \text{k}\Omega$, $R_{\text{bottom}} = 4.7\ \text{k}\Omega$.
- $R_{\text{th}} \approx 4.59\ \text{k}\Omega$
- $C_{\text{filter}} = 10\ \text{nF}$
- $f_c = \frac{1}{2\pi\times4590\times10\times10^{-9}} \approx 3.47\ \text{kHz}$
- *Evaluation*: Balanced for high-speed tracking and noise rejection.

7. Current Sense (CC6937)

- **Type**: Isolated Hall Effect.
- **Sensitivity**: Check variant (e.g., 66mV/A for 20A).
- **Range**: ±20A (Full scale ≈ 2.97V on 3.3V ADC).
- **VREF**: 100nF bypass to GND required.

8. 200kHz Migration: Component Re-Check

This section re-verifies the parts already selected for 100kHz operation against a doubled switching frequency, using values pulled directly from the vendored datasheets (not assumed). Nothing here has been implemented in firmware or hardware yet — this is the pre-check called for by the `STANDARDS.md` "Part selection & calculations" mandate before the `ROADMAP_V1.3.md` Phase 4 checklist item is closed.

8.1. Main Switching MOSFETs (Q1-Q4, BSC030N08NS5, C501507)

Source: Infineon `BSC030N08NS5` datasheet, Rev. 2.3, 2019-10-31.

Parameter	Value	Condition
$R_{\mathrm{DS(on)}}$	2.6 mΩ typ / 3.0 mΩ max	V_{GS}=10V, I_D=50A
Q_g (total gate charge)	61 nC typ / 76 nC max	V_{DD}=40V, I_D=50A, V_{GS}=0→10V
t_r (turn-on rise)	12 ns typ	V_{DD}=40V, V_{GS}=10V, I_D=50A, $R_{\mathrm{G,ext}}$ =3Ω
t_f (turn-off fall)	13 ns typ	same
C_{oss}	700 pF typ / 910 pF max	V_{DS}=40V, f=1MHz
$R_{\mathrm{th\ JA}}$ (PCB only, no heatsink)	50 K/W	6 cm² copper, 1-layer, natural convection
$T_{\mathrm{j,max}}$	150 °C	—

Conduction loss (frequency-independent):

$$P_{\mathrm{cond}} = I_{\mathrm{rms}}^2 \times R_{\mathrm{DS(on),max}} = 20^2 \times 0.0030 \approx * 1.2 \text{ W} *$$

Switching loss (linear-approximation model, worst case $V_{\mathrm{DS}} = 80\mathrm{V}$, $I_D = 20\mathrm{A}$):

$$P_{\mathrm{sw}} = \frac{1}{2} \times V_{\mathrm{DS}} \times I_D \times (t_r + t_f) \times f_{\mathrm{sw}}$$

	100 kHz (current)	200 kHz (proposed)
P_{sw} per hard-switched MOSFET	2.0 W	4.0 W
P_{cond} per MOSFET	1.2 W	1.2 W (unchanged)
Total est. loss	**≈3.2 W**	**≈5.2 W**

Thermal flag. With PCB-only cooling ($R_{\mathrm{th\ JA}}$=50 K/W, no heatsink/thermal vias), the datasheet's own max package dissipation at T_A=25°C is $P_{\mathrm{tot}} = \frac{150-25}{50} = * 2.5 \text{ W} *$ — already below the ≈3.2W estimate at **100kHz**, let alone ≈5.2W at 200kHz. This assumes worst-case V_{DS}/ I_D coincidence and a simplified switching-loss model (real loss depends on which of Q1-Q4 is hard-switched vs. synchronous-rectifying in the actual buck-boost topology, and gate resistance/ drive strength affect t_r/t_f in-circuit). This is exactly what `ROADMAP_V1.3.md` Phase 3′s open item ("20A Thermal Design: widen high-power traces, implement massive thermal via grid") already anticipates — that item should land **before** or **alongside** the 200kHz switch, not after. The bottom-side $R_{\mathrm{th\ JC}}$ path (0.5-0.9 K/W typ/max) is dramatically better than the PCB-only path and is what a proper thermal via array would unlock.

Gate drive loss (in the driver/gate resistor, not the MOSFET junction — informational):

$$P_{\mathrm{gate}} = Q_g \times V_{\mathrm{GS}} \times f_{\mathrm{sw}} = 76 \times 10^{-9} \times 10 \times f_{\mathrm{sw}}$$

At 100kHz: 76mW/MOSFET. At 200kHz: 152mW/MOSFET. Not a concern for the driver IC's thermal budget either way.

8.2. Gate Driver (IRS21867STRPBF, C52290)

Source: Infineon `IRS21867S` datasheet v01.00.

Parameter	Value	Condition
t_{on} / t_{off} propagation delay	170 ns typ / 250 ns max	$V_{\mathrm{CC}} = V_{\mathrm{BS}}$=15V, C_L=1000pF

Parameter	Value	Condition
MT (delay matching, $\|t_{\text{on}} - t_{\text{off}}\|$)	35 ns max	—
t_r / t_f (driver output)	22/18 ns typ, 38/30 ns max	—
Output source/sink current	4.0 A typ	—

At 200kHz the switching period is 5.0µs vs. 10µs at 100kHz. The 170-250ns propagation delay grows from 2.5% to 5% of the period — worth including explicitly in dead-time budgeting, but well within normal margins; **not** a hard blocker on 200kHz operation. Output drive current (4.0A) is unchanged and still comfortably swings Q_g=76nC in the tens-of-ns range required. No datasheet-stated maximum switching frequency applies here.

8.3. Main Inductor (currently L4, FC-SE2822-150M, C46553544)

Re-deriving Section 4′s formula at 200kHz:

$$L_{\text{buck}}(200\text{kHz}) = \frac{40 \times 40}{4.0 \times 200000 \times 80} = *\,25.0\mu\text{H}\,*$$

This **halves** the 100kHz target (50µH → 25µH), which also shifts the "target range" from 33-47µH down toward roughly **16.5-23.5µH** if the same 20%-ripple-target methodology and 25A I_{sat} margin are kept.

> **Ties back to the open L4 question in `STANDARDS.md`.** L4 (`FC-SE2822-150M`) was flagged there because its part-name convention implies ≈15µH, well under the 33-47µH target for **100kHz**. At **200kHz**, that same 15µH is much closer to (if still slightly under) the new ≈16.5-23.5µH target — the frequency change would substantially close that gap rather than widen it. This is **not** confirmation the two were chosen together on purpose; it's a reason to verify the inductance value from the datasheet/markings rather than resolve it by coincidence. I_{sat} still needs independent confirmation against the 20-25A requirement regardless of frequency.

8.4. Summary

Item	Verdict at 200kHz
MOSFETs (BSC030N08NS5)	Electrically fine (Vds/Rds(on)/Qg headroom unchanged), but switching loss ≈doubles — thermal design (ROADMAP Phase 3) must land first
Gate driver (IRS21867STRPBF)	No concern — propagation delay and drive current have ample margin
Main inductor (L4)	Required inductance halves to ≈25µH — worth re-deriving the target range and confirming L4′s actual value/Isat before committing
Bulk caps / TVS / voltage sensing	Frequency-independent, no re-check needed